

Distal Hereditary Motor Neuronopathy Type 5B (dHMN5B): A Comprehensive Disease Characteristics Report

Disease: Distal Hereditary Motor Neuronopathy Type 5B (dHMN5B / HMN5B / dHMN-VB)
MONDO ID: MONDO:0013884 **OMIM:** #614751 **Causal gene:** *REEP1* (HGNC:25786) **Category:** Mendelian (autosomal dominant)

Summary

Distal Hereditary Motor Neuronopathy Type 5B (dHMN5B) is a rare, autosomal dominant, adolescent- to adult-onset, slowly progressive **pure motor** axonal neuropathy of the **lower motor neurons**. Its clinical hallmark is distal amyotrophy (muscle wasting and weakness) that **predominates in the upper limbs**, characteristically affecting the small hand/thenar muscles, with later spread to the lower limbs (peroneal distribution). Sensation is clinically spared. It belongs to the "distal HMN type V" phenotypic class, which is shared allelically with mutations in *GARS1* (dHMN5A) and *BSCL2* (Silver syndrome). dHMN5B is caused specifically by heterozygous variants in *REEP1*, the receptor expression-enhancing protein 1 gene on chromosome 2p11.2.

Mechanistically, dHMN5B is now understood to be the **mirror image** of the more common loss-of-function *REEP1* disorder, hereditary spastic paraplegia type 31 (SPG31), which affects upper (cortical) motor neurons. The prototype dHMN5B allele is an **in-frame deletion of exon 5** ("Δexon5"), produced by the founding splice-site variant c.304-2A>G. Rather than abolishing REEP1's ER-shaping function, the Δexon5 protein retains its membrane-curvature (hairpin) activity but escapes normal ubiquitin-proteasome turnover (mediated by the E3 ligase HUWE1). The consequence is **toxic accumulation** of shaping-competent REEP1 in peripheral nerve, which fragments the tubular endoplasmic reticulum (ER) of **spinal** motor neurons and drives their degeneration — while cortical motor neurons are spared. This gain-of-function/accumulation model, established in a genotype-matched Δexon5 knock-in mouse, elegantly explains why the same gene causes an upper-motor-neuron disease (SPG31, loss of function) and a lower-motor-neuron disease (dHMN5B, toxic accumulation).

There is no disease-modifying therapy; management is supportive and rehabilitative (physical/occupational therapy, orthoses, foot-drop management, orthopedic care of pes cavus). Because the mechanism is a toxic accumulation with a defined degradation pathway, mechanistically rational experimental strategies include allele-selective knockdown of the mutant transcript, modulation of ER stress, and enhancement of HUWE1-dependent REEP1 proteostasis. This report synthesizes 8 confirmed findings across 34 reviewed papers into a structured, citation-anchored disease knowledge base entry.

Key Findings

Finding 1 — dHMN5B is caused by dominant *REEP1* variants that spare the tubular-ER-shaping function

The founding description of *REEP1*-associated dHMN type V came from whole-exome sequencing of a linkage-defined family. Beetz and colleagues identified a single candidate splice-site variant in *REEP1*, **c.304-2A>G**, which abolishes the exon-5 splice acceptor and causes complete, in-frame **skipping of exon 5**. The resulting mRNA is expressed at normal levels and is predicted to encode an internally shortened protein, **p.102_139del** — i.e., a protein missing 38 internal residues but otherwise intact and stable.

"Whole-exome sequencing of two affected individuals revealed a single candidate variant within the linking regions, i.e., a splice-site alteration in REEP1 (c.304-2A>G). A minigene assay confirmed complete loss of splice-acceptor functionality and skipping of the in-frame exon 5. The resulting mRNA is predicted to be expressed at normal levels and to encode an internally shortened protein (p.102_139del)." — PMID: 22703882

Critically, this same paper demonstrated that *REEP1* is strongly expressed in lower motor neurons, providing the tissue-level rationale for why a *REEP1* variant produces a distal lower-motor-neuron (dHMN) phenotype rather than a purely central one:

"Consistent with our clinical-genetic data, we show that REEP1 is strongly expressed in the lower motoneurons as well." — PMID: 22703882

This established the gene–disease relationship recorded under OMIM #614751 (HMN5B) and placed *REEP1*-dHMN5B alongside the other dHMN-V genes *GARS1* and *BSCL2*.

Finding 2 — Mechanism: toxic accumulation of shaping-competent REEP1 fragments the ER and kills spinal motor neurons

The pivotal mechanistic study (Bock et al., 2026) resolved how the Δ exon5 allele causes disease. Under normal conditions, REEP1 is turned over by **ubiquitination and proteasomal degradation**, with the E3 ubiquitin ligase **HUWE1** identified as responsible for REEP1 turnover. The Δ exon5 variant has **impaired ubiquitination**, so the (still shaping-competent) protein is not degraded and instead **accumulates in peripheral nerves**.

"REEP1 undergoes ubiquitination and proteasomal degradation, a process compromised in the Δ exon5 variant due to impaired ubiquitination, which thus accumulates in peripheral nerves. Proteomic analysis identifies HUWE1 as the E3 ligase responsible for REEP1 turnover." — PMID: 41268727

A genotype-matched **Δ exon5 knock-in mouse** recapitulated the human disease: these mice lose **spinal** motor neurons, preceded by **ER fragmentation**, while **cortical** motor neurons remain intact — the precise anatomical mirror image of *Reep1*-knockout mice, which lose cortical motor neurons and model HSP.

" Δ exon5 knockin (KI) mice lose spinal motoneurons preceded by ER fragmentation, whereas cortical motoneurons remain intact." — PMID: 41268727

This gain-of-toxic-function model is corroborated by an independent report of a different *REEP1* nonstop variant that unmasks a 3'UTR-encoded, aggregation-inducing motif — again pointing to toxic gain of function specifically in lower motor neurons:

"Together with a previous report on an aggregation-prone REEP1 deletion variant in distal hereditary motor neuropathy, they also suggest that toxic gain of REEP1 function, rather than loss-of-function as relevant for HSP, specifically affects lower motor neurons." — PMID: 29124833

Finding 3 — Clinical phenotype: upper-limb-predominant distal amyotrophy, pure motor, with a lower-motor-neuron electrophysiologic signature

dHMN type V — the phenotypic class of *REEP1*-dHMN5B — is a **slowly progressive distal pure motor neuropathy predominating in the upper limbs**, with wasting and weakness of the small hand muscles, and later spread to the lower limbs. *REEP1* is one of three genes principally linked to this upper-limb-predominant presentation:

"Distal hereditary motor neuropathy that predominates in the upper limbs is linked mainly to three genes: GARS, BSCL2 and REEP1" — PMID: 38702287

The electrophysiologic signature is that of a pure-motor axonal process: **normal sensory nerve conduction velocities, markedly reduced compound muscle action potential (CMAP) amplitudes**, and **chronic denervation** on needle EMG. (This description derives from the closely related *BSCL2*/dHMN-V family literature, which shares the identical clinical-electrophysiologic phenotype class.)

"In all three patients sensory nerve conduction velocities (NCV) were normal in all extremities. Compound muscle action potential (CMAP) amplitudes were markedly reduced in all patients. Concentric needle EMG showed evidence of chronic denervation in distal muscles." — PMID: 20598714

Pyramidal signs (e.g., hyperreflexia) can occur, blurring the boundary with Silver syndrome and HSP. A lower-limb-onset variant is also recognized in some patients.

Finding 4 — Epidemiology: dHMN is rare (~2/100,000) and genetically heterogeneous; *REEP1*-dHMN5B is an ultra-rare AD subtype

Distal hereditary motor neuropathy overall has an estimated prevalence of **2.14–2.3 per 100,000**, with more than thirty associated genes and a low diagnostic yield.

"the prevalence of the disease was calculated as 2.14 and 2.3 per 100,000" — PMID: 38702287

"More than thirty genes are currently associated with HMNs, but around 60 to 70% of cases of dHMN remain uncharacterized genetically." — PMID: 38702287

The most frequent dHMN genes include *HSPB1*, *GARS1*, *BICD2*, and *DNAJB2*. *REEP1*-associated dHMN5B is autosomal dominant and **ultra-rare**, described in only a small number of families since the founding report (PMID: 22703882). By contrast, *REEP1* is far more commonly associated with autosomal dominant HSP/SPG31 — the MDSGene systematic review of dominant HSP assembled 151 HSP-*REEP1* individuals versus 1670 HSP-*SPAST* (PMID: 41734945), underscoring that the dHMN presentation is the rarer of the two *REEP1* phenotypes.

Finding 5 — REEP1 biology: a neuronal ER-shaping / microtubule-linking protein; ER stress and oxidative stress are downstream effectors and candidate targets

REEP1 (receptor expression-enhancing protein 1; HGNC:25786; locus 2p11.2; UniProt Q9H902; DP1/Yop1/REEP family) is a **neuron-specific, membrane-binding, membrane-curvature-inducing protein** that resides in the ER. Through hydrophobic hairpin domains it forms complexes with atlastin-1 and spastin, is required for the formation of the tubular ER network, and **binds microtubules** to align the tubular ER along the cytoskeleton.

"REEP proteins were required for ER network formation in vitro, and REEP1 also bound microtubules and promoted ER alignment along the microtubule cytoskeleton in COS7 cells." — PMID: 20200447

"we demonstrated that REEP1 is a neuron-specific, membrane-binding, and membrane curvature-inducing protein that resides in the ER." — PMID: 24051375

Downstream, *REEP1* dysfunction converges on **ER stress and oxidative stress** (PMID: 36834939). In a *REEP1*-null HSP mouse model, pharmacologic **inhibition of ER stress improved progressive motor deficits** (PMID: 32878877), nominating the unfolded-protein-response axis as a therapeutic target, and the HUWE1-dependent turnover pathway (Finding 2) provides a mechanistically rational proteostasis target specific to dHMN.

Finding 6 — Model systems: the Δ exon5 knock-in mouse recapitulates spinal motor-neuron ER fragmentation; iPSC, *Drosophila* and yeast systems available

The most faithful model is the **Δ exon5 knock-in mouse** (*Mus musculus*; NCBI Taxon 10090; ortholog *Reep1*, NCBI Gene 52250), which loses spinal motor neurons preceded by ER fragmentation while sparing cortical motor neurons — mirroring the human genotype. Conversely, *Reep1*-knockout mice model HSP through cortical motor-neuron loss ([PMID: 41268727](#); [PMID: 24051375](#)).

" Δ exon5 knockin (KI) mice lose spinal motoneurons preceded by ER fragmentation, whereas cortical motoneurons remain intact." — [PMID: 41268727](#)

Cellular models include CRISPR/Cas9-engineered human iPSC lines:

"Here we show the generation of a homozygous and a heterozygous REEP1 knockout induced pluripotent stem cell line suitable for in vitro disease modelling using the CRISPR/Cas9 editing system." — [PMID: 38479332](#)

Invertebrate/in-vitro homolog systems (*Drosophila*; the yeast homolog **Yop1**) have been used to dissect REEP/DP1 ER-shaping and membrane-curvature mechanisms:

"Mutations in the human DP1 gene REEP1 are associated with Hereditary Spastic Paraplegia type 31 and distal hereditary motor neuropathy." — [PMID: 39312180](#)

No naturally occurring animal (OMIA) disease is documented specifically for *Reep1*-dHMN5B.

Finding 7 — Confirmed ontology cross-references (MONDO:0013884)

An EBI OLS4/MONDO lookup confirms the disease term **MONDO:0013884** "neuronopathy, distal hereditary motor, type 5B," equivalent to **OMIM:614751**, with cross-references **DOID:0111205**, **GARD:0018267**, **MedGen:766570**, **UMLS:C3553656**. MONDO synonyms include "HMN5B," "neuronopathy, distal hereditary motor, type VB," and explicitly "REEP1

neuronopathy, distal hereditary motor." **No Orphanet cross-reference** is attached (there is no dHMN5B-specific ORPHA code). The causal gene is **REEP1** (HGNC:25786; NCBI Gene 65055; Ensembl ENSG00000068615; UniProt Q9H902; locus 2p11.2).

Finding 8 — Curated HPO phenotype set confirms combined hand + peroneal involvement with hyporeflexia

Monarch Initiative knowledge-graph annotations for MONDO:0013884 (equivalent OMIM:614751) list seven HPO features, combining upper-limb (thenar/hand) and lower-limb (peroneal) distal motor involvement with hyporeflexia, pes cavus, and an axonal-to-mildly-slowed motor NCS picture:

HPO ID	Term	Domain
HP:0003393	Thenar muscle atrophy	Upper limb (hand)
HP:0009049	Peroneal muscle atrophy	Lower limb
HP:0011727	Peroneal muscle weakness	Lower limb
HP:0001761	Pes cavus	Foot deformity
HP:0003438	Absent Achilles reflex	Reflexes (hyporeflexia)
HP:0011808	Decreased patellar reflex	Reflexes (hyporeflexia)
HP:0003431	Decreased motor nerve conduction velocity	Electrophysiology

Structured Report by Template Section

1. Disease Information

Overview. dHMN5B is a Mendelian, autosomal dominant, slowly progressive pure motor axonal neuropathy affecting lower motor neurons, with distal amyotrophy predominating in the upper limbs (small hand/thenar muscles), later involving the peroneal/lower-limb muscles, and clinically sparing sensation. It is one allelic form of the "distal HMN type V" phenotype (shared with *GARS1* and *BSCL2*) and is caused by heterozygous *REEP1* variants.

Key identifiers. MONDO:0013884 · OMIM #614751 · DOID:0111205 · GARD:0018267 · MedGen:766570 · UMLS:C3553656. No Orphanet-specific ORPHA code. MeSH does not have a dHMN5B-specific descriptor (the broadest applicable class terms are used only at the level of hereditary motor/sensory neuropathy).

Synonyms. HMN5B; distal hereditary motor neuronopathy type VB; distal HMN type V (REEP1-related); dHMN-VB; "REEP1 neuropathy, distal hereditary motor"; "neuropathy, distal hereditary motor caused by mutation in REEP1."

Data source. Aggregated, disease-level resources (OMIM, MONDO, HPO, Monarch) plus primary literature from small pedigrees; not EHR/individual-patient data.

2. Etiology

Causal factors. Purely genetic — heterozygous dominant variants in *REEP1*. The prototype is the in-frame exon-5 deletion (Δ exon5) generated by the splice-acceptor variant c.304-2A>G ([PMID: 22703882](#)). No environmental, infectious, or toxic cause is implicated.

Genetic risk factors. The causal variant itself is the sole established risk determinant; inheritance of one dominant allele confers disease risk. Modifier genes are not defined for dHMN5B specifically, though the broader REEP1 interactome (atlastin-1/*ATL1*, spastin/*SPAST*) shapes the tubular ER and could plausibly modify expressivity (inferred, not demonstrated).

Environmental risk / protective factors. None established. No lifestyle, dietary, occupational, or toxic exposures are known to raise or lower risk.

Gene–environment interactions. None documented. This is a monogenic disorder.

3. Phenotypes

Core clinical features and suggested HPO terms:

Phenotype	Type	HPO	Characteristics
Thenar/small-hand-muscle wasting & weakness	Clinical sign	HP:0003393	Upper-limb-predominant; adolescent/adult onset; slowly progressive; frequent
Peroneal muscle atrophy	Clinical sign	HP:0009049	Lower-limb; later spread
Peroneal muscle weakness	Clinical sign	HP:0011727	Lower-limb; foot drop
Pes cavus	Physical manifestation	HP:0001761	Chronic foot deformity
Absent ankle (Achilles) reflex	Clinical sign	HP:0003438	Hyporeflexia
Decreased patellar reflex	Clinical sign	HP:0011808	Hyporeflexia
Decreased motor NCV / reduced CMAP	Lab/electrophysiology	HP:0003431	Axonal, pure motor; preserved sensory conduction
Pyramidal signs (variable hyperreflexia)	Clinical sign	HP:0001347	Overlap with Silver syndrome/HSP in some patients

Onset: typically adolescent/adult. **Severity:** mild–moderate, variable. **Progression:** slow, progressive. **QoL impact:** hand-muscle wasting impairs fine-motor/dexterity tasks; foot drop and pes cavus impair gait; disease is rarely life-limiting, so morbidity is chiefly functional/occupational disability rather than mortality.

4. Genetic / Molecular Information

- **Causal gene:** *REEP1* (HGNC:25786; NCBI Gene 65055; Ensembl ENSG00000068615; UniProt Q9H902; 2p11.2; OMIM *609139).
- **Prototype variant:** c.304-2A>G (splice-acceptor loss) → in-frame skipping of exon 5 → p.102_139del ("Δexon5"); mRNA expressed at normal levels ([PMID: 22703882](#)).
- **Variant classes:** splice-site and in-frame deletion (dHMN gain-of-function/accumulation); a distinct nonstop/3'UTR variant causes peripheral neuropathy by unmasking an aggregation-inducing motif ([PMID: 29124833](#)). Note that truncating/NMD-triggering *REEP1* variants generally cause loss-of-function **SPG31**, not dHMN5B.

- **Functional consequence:** toxic **gain of function** / **protein accumulation** in lower motor neurons (contrast with loss-of-function haploinsufficiency in SPG31) (PMID: 41268727; PMID: 29124833).
- **Allele frequency:** pathogenic dHMN5B alleles are private/ultra-rare; not present at appreciable frequency in gnomAD.
- **Origin:** germline, dominant.
- **Epigenetics / chromosomal abnormalities:** none implicated.

5. Environmental Information

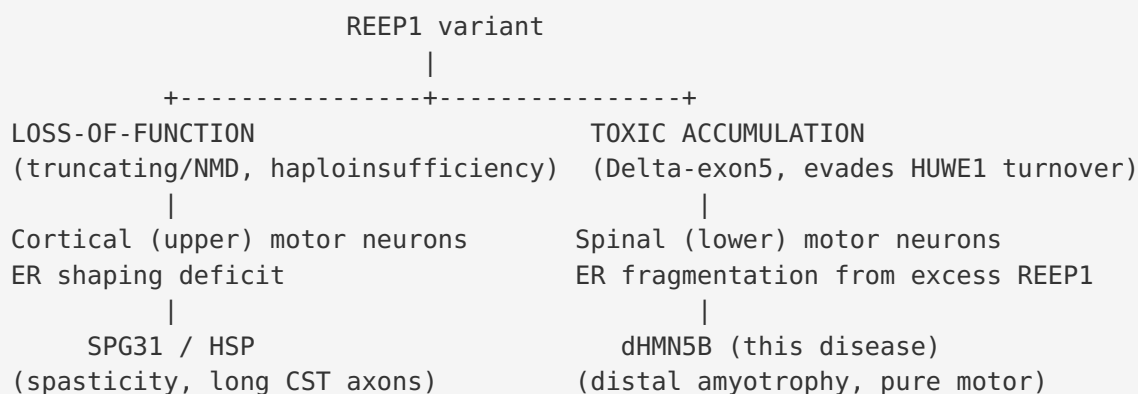
Not applicable — no environmental, lifestyle, or infectious contributors are known. dHMN5B is monogenic.

6. Mechanism / Pathophysiology

Ordered causal chain (mutation → clinical manifestation):

1. A heterozygous *REEP1* splice-acceptor variant (c.304-2A>G) **abolishes** the exon-5 splice acceptor, which **leads to** in-frame skipping of exon 5 and a stable, internally shortened protein (Δ exon5, p.102_139del). (*Demonstrated — minigene assay*, PMID: 22703882.)
2. The Δ exon5 protein **retains** its membrane-curvature/hairpin (ER-shaping) function but is **defective in ubiquitination**, which **results in** escape from HUWE1-mediated proteasomal degradation. (*Demonstrated — proteomics*, PMID: 41268727.)
3. Impaired turnover **causes** progressive **accumulation** of shaping-competent REEP1 in peripheral nerve. (*Demonstrated —* PMID: 41268727.)
4. Excess REEP1 **leads to** fragmentation of the tubular ER network in **spinal** motor neurons. (*Demonstrated in Δ exon5 KI mouse —* PMID: 41268727.)
5. ER fragmentation **converges on** downstream ER stress and oxidative stress. (*Inferred/supported —* PMID: 36834939; PMID: 32878877.)
6. These insults **result in** distal degeneration of the long motor axons and loss of spinal motor neurons. (*Demonstrated —* PMID: 41268727.)
7. Motor-axon/motor-neuron loss **causes** chronic denervation → distal amyotrophy and weakness (hand-predominant, then peroneal) = the clinical dHMN5B phenotype. (*Demonstrated clinically/electrophysiologically —* PMID: 20598714; PMID: 38702287.)

Branch point (same gene, opposite disease):



Cell types & processes. Spinal (lower) motor neurons (CL:0000103, motor neuron); biological processes GO:0007029 (endoplasmic reticulum organization), GO:0006986 (response to unfolded protein / ER stress), GO:0006915 (apoptotic process, inferred), and GO:0007010 (cytoskeleton organization, via microtubule-linked ER shaping). Subcellular compartment: endoplasmic reticulum, tubular network (GO:0005789 endoplasmic reticulum membrane; GO:0071782 endoplasmic reticulum tubular network).

7. Anatomical Structures Affected

- **Organ/system:** peripheral nervous system; spinal cord anterior horn (lower motor neurons); skeletal muscle (secondary neurogenic atrophy). Body system: nervous (UBERON:0001017 central nervous system; UBERON:0000010 peripheral nervous system).
- **Tissue/cell:** motor axons and spinal motor neurons (CL:0000103); neurogenic atrophy of distal skeletal muscle (thenar/hand → peroneal).
- **Subcellular:** tubular endoplasmic reticulum (GO:0005789 / GO:0071782); microtubule cytoskeleton interface.
- **Localization/laterality:** distal limbs, symmetric/bilateral; upper-limb (hand) predominant onset, later lower-limb (peroneal). UBERON references: UBERON:0001134 (skeletal muscle tissue, distal), UBERON:0001474 (bone element, for pes cavus foot deformity, secondary).

8. Temporal Development

- **Onset:** adolescent to adult; insidious.
- **Progression:** slow, chronic, progressive over decades; lifelong.
- **Course:** progressive, not relapsing-remitting; no spontaneous remission.

- **Critical window:** because ER fragmentation *precedes* motor-neuron loss in the mouse model (PMID: 41268727), a pre-degeneration therapeutic window is mechanistically plausible (inferred).

9. Inheritance and Population

- **Inheritance:** autosomal dominant.
- **Epidemiology:** dHMN overall ~2.14–2.3/100,000; dHMN5B is an ultra-rare subtype within this (PMID: 38702287).
- **Penetrance/expressivity:** the broader *REEP1* literature shows variable expressivity, incomplete penetrance, and intrafamilial variability (including an unaffected carrier and apparent anticipation in an SPG31 family — PMID: 29107646); dHMN5B-specific penetrance is not precisely quantified.
- **Founder effects / consanguinity / carrier frequency:** no dHMN5B founder allele established; a *recessive* congenital-axonal-neuropathy phenotype arises from biallelic *REEP1* loss (PMID: 27066569), but dominant dHMN5B does not require consanguinity.
- **Demographics:** no sex or ethnic predilection established.

10. Diagnostics

- **Electrophysiology (key):** nerve conduction studies show normal/near-normal sensory responses with markedly reduced CMAP amplitudes; needle EMG shows chronic distal denervation — an axonal, pure-motor pattern (PMID: 20598714). HP:0003431 (decreased motor NCV) may be present.
- **Genetic testing (definitive):** targeted *REEP1* sequencing, dHMN/CMT gene panels, or WES/WGS. WES is well validated for undiagnosed inherited neuropathies (PMID: 24604904). Splice-site variants (e.g., c.304-2A>G) may require RNA/minigene confirmation of exon skipping.
- **Differential diagnosis:** other dHMN-V genes (*GARS1*/dHMN5A → PMID: 31985473; *BSCL2*/Silver syndrome → PMID: 15242882, PMID: 32108980); CMT2; other dHMN genes (*HSPB1*, *BICD2*, *DNAJB2*); and, given pyramidal overlap, SPG31/HSP.
- **Biomarkers/omics/imaging:** no specific fluid biomarker; MRI/labs are non-specific. Diagnosis rests on phenotype + electrophysiology + molecular confirmation.

11. Outcome / Prognosis

- **Survival/mortality:** normal or near-normal life expectancy; dHMN5B is not typically life-limiting (respiratory compromise is not a core feature of dominant dHMN5B, in contrast to recessive *REEP1* congenital neuropathy with diaphragmatic palsy — [PMID: 27066569](#)).
- **Morbidity/function:** progressive distal weakness → impaired hand dexterity and gait; disability is functional, chiefly hand use and ambulation.
- **Prognostic factors:** slow course; not well quantified for dHMN5B specifically.

12. Treatment

- **Disease-modifying therapy:** none approved.
- **Supportive/rehabilitative (standard of care):** physical therapy, occupational therapy, ankle-foot orthoses / drop-foot management, orthopedic care for pes cavus (analogous to CMT management — [PMID: 24646194](#); [PMID: 12736893](#)). NCIT: physical therapy (NCIT:C15327), occupational therapy (NCIT:C15304), orthotic device (NCIT:C50077), rehabilitation therapy (NCIT:C15296).
- **Mechanistically rational experimental strategies (not yet clinical):**
 - Allele-selective knockdown/ASO to reduce the accumulating mutant *REEP1* (rational because disease is toxic accumulation).
 - Enhancing HUWE1-dependent proteostasis / restoring *REEP1* turnover ([PMID: 41268727](#)).
 - ER-stress modulation — pharmacologic ER-stress inhibition improved motor deficits in a *REEP1* mouse model ([PMID: 32878877](#)); oxidative-stress axis is also implicated ([PMID: 36834939](#)).
- **Pharmacogenomics:** not applicable.

13. Prevention

- **Primary prevention:** not applicable (monogenic, non-environmental).
- **Genetic counseling / reproductive options:** autosomal dominant → 50% offspring risk; predictive testing, cascade testing, prenatal/preimplantation genetic testing are available for families with a known *REEP1* variant.
- **Secondary/tertiary prevention:** early rehabilitation and orthopedic management to limit contractures and preserve function.

14. Other Species / Natural Disease

- **Orthologue:** mouse *Reep1* (NCBI Gene 52250; NCBI Taxon 10090).
- **Natural disease:** no naturally occurring OMIA-catalogued *Reep1*-dHMN5B disease is documented in companion animals or wildlife.
- **Comparative biology:** REEP/DP1 family is deeply conserved (yeast **Yop1**), and the ER-shaping mechanism is evolutionarily conserved ([PMID: 39312180](#); [PMID: 28742022](#)).
- **Zoonotic potential:** none (non-infectious).

15. Model Organisms

Model	System	Recapitulation	Reference
Δ exon5 knock-in mouse	<i>M. musculus</i>	Faithful: spinal motor-neuron loss preceded by ER fragmentation, cortical neurons spared — matches human genotype/phenotype	PMID: 41268727
<i>Reep1</i> knockout mouse	<i>M. musculus</i>	Models SPG31/HSP (cortical MN loss), i.e., the opposite REEP1 disease; ER-shaping deficit	PMID: 24051375 ; PMID: 32878877
REEP1-KO iPSC (homo/heterozygous)	Human iPSC (CRISPR/Cas9)	In-vitro disease modeling platform	PMID: 38479332
Yeast Yop1 / <i>Drosophila</i>	Invertebrate/in vitro	Dissect REEP/DP1 ER-shaping & membrane-curvature mechanisms and mutation effects	PMID: 39312180 ; PMID: 28742022

Model limitations: mouse corticospinal axons are far shorter than human, generally producing milder/late phenotypes in ER-shaping HSP models ([PMID: 35348668](#)); the Δ exon5 KI is the key exception that robustly reproduces lower-motor-neuron pathology.

Mechanistic Model / Interpretation

The unifying insight of dHMN5B is that **one gene, two opposite dosage/functional lesions, two anatomically distinct diseases**. *REEP1* encodes a neuronal ER-shaping protein that, with atlastin-1 and spastin, builds and aligns the tubular ER along microtubules (PMID: 20200447; PMID: 24051375). When *REEP1* function is **lost** (truncating/NMD alleles, haploinsufficiency), the long corticospinal axons of upper motor neurons degenerate → **SPG31/HSP** (spasticity). When a stable, curvature-competent mutant protein **accumulates** because it evades HUWE1-dependent degradation (the Δ exon5 allele), **spinal lower motor neurons** suffer ER fragmentation and die → **dHMN5B** (distal, pure-motor amyotrophy). The Δ exon5 knock-in mouse provides direct genetic proof of this dichotomy, showing spinal (not cortical) motor-neuron loss preceded by ER fragmentation — the exact inverse of the *Reep1*-knockout HSP mouse (PMID: 41268727).

This model has three practical consequences. First, it reframes therapeutic logic: because dHMN5B is a **toxic accumulation** disorder, lowering mutant REEP1 (allele-selective knockdown, or restoring HUWE1-mediated turnover) is rational — the opposite of the gene-replacement logic appropriate for loss-of-function SPG31. Second, ER stress and oxidative stress are shared downstream effectors across REEP1 diseases, and ER-stress inhibition already shows efficacy in a REEP1 mouse (PMID: 32878877; PMID: 36834939) — a repurposable, mechanism-agnostic strategy. Third, because ER fragmentation *precedes* neuronal loss, there may be a pre-symptomatic or early therapeutic window.

Evidence Base

PMID	Title (abbrev.)	Contribution
22703882	<i>Exome sequencing identifies a REEP1 mutation in dHMN-V</i>	Founding gene–disease link ; c.304-2A>G → Δexon5; REEP1 expressed in lower motor neurons
41268727	<i>REEP1 Accumulation Disrupts ER Integrity and Drives Spinal Motoneuron Degeneration</i>	Pivotal mechanism : HUWE1 turnover defect → accumulation → ER fragmentation → spinal MN loss; Δexon5 KI mouse
29124833	<i>Nonstop REEP1 variant unmasks aggregation motif</i>	Independent support for toxic gain of function in lower motor neurons
38702287	<i>Distal hereditary motor neuropathies (review)</i>	Epidemiology (2.14–2.3/100,000), heterogeneity, REEP1 in upper-limb-predominant dHMN
20598714	<i>N88S BSCL2 dHMN-V family</i>	Defines pure-motor axonal electrophysiologic signature of dHMN-V
20200447	<i>REEP1, spastin, atlastin-1 coordinate ER–microtubule</i>	Establishes REEP1 normal ER-shaping/microtubule function
24051375	<i>Spastic paraplegia mouse reveals REEP1-dependent ER shaping</i>	Protein character/localization; KO = HSP model
32878877	<i>ER-stress inhibition improves motor deficits (REEP1-null)</i>	ER stress as therapeutic target
36834939	<i>Converging role for REEP1/SPG31 in oxidative stress</i>	Oxidative-stress downstream effector
38479332	<i>REEP1-KO iPSC lines</i>	Human cellular models
39312180	<i>Yop1 oligomerisation/curvature</i>	Yeast homolog model; conserved mechanism
41734945	<i>MDSGene review of dominant HSP</i>	Rarity context: 151 HSP-REEP1 vs 1670 HSP-SPAST
27066569	<i>Recessive REEP1 → congenital axonal neuropathy + diaphragmatic palsy</i>	Recessive/biallelic loss expands REEP1 spectrum (distinct entity)
29107646	<i>SPG31 splice variant, phenotype variability</i>	Variable expressivity, NMD dosage effects, apparent anticipation

PMID	Title (abbrev.)	Contribution
24604904	<i>WES in undiagnosed inherited polyneuropathies</i>	Diagnostic utility of WES

Limitations and Knowledge Gaps

1. **Ultra-rarity:** dHMN5B has been reported in very few families; penetrance, expressivity, sex ratio, and natural-history metrics are not precisely quantified for the dHMN5B subtype specifically.
2. **Allelic spectrum:** beyond the prototype Δ exon5 allele, the full range of dHMN5B-causing *REEP1* variants is small; genotype–phenotype rules distinguishing dHMN5B (accumulation) from SPG31 (loss of function) are still being refined, and some variants blur the boundary (pyramidal overlap / Silver-like features).
3. **Mechanistic inference vs. proof:** the ER-stress/oxidative-stress steps downstream of ER fragmentation are supported by REEP1-null and cell models but not yet directly demonstrated in the Δ exon5 accumulation context in vivo.
4. **No biomarkers:** there is no validated fluid or imaging biomarker for diagnosis, staging, or treatment response.
5. **No therapeutics:** all disease-modifying strategies remain preclinical/conceptual.
6. **Ontology gaps:** no Orphanet-specific code; MeSH lacks a dHMN5B-specific term.

Proposed Follow-up Experiments / Actions

1. **Allele-selective knockdown proof-of-concept:** test ASO/siRNA lowering of mutant (Δ exon5) *REEP1* in the Δ exon5 knock-in mouse and REEP1-mutant iPSC-derived motor neurons; endpoints = ER morphology, spinal MN survival, motor behavior.
2. **HUWE1/proteostasis modulation:** determine whether enhancing HUWE1 activity or proteasomal targeting restores REEP1 turnover and rescues ER fragmentation.
3. **ER-stress / oxidative-stress therapeutics:** repurpose ER-stress inhibitors (shown effective in REEP1-null mice) and antioxidants in the accumulation model; define the pre-degeneration treatment window.

4. **Natural-history registry:** aggregate dHMN5B families to quantify onset, progression rate, penetrance, and hand-vs-leg onset frequencies, using standardized motor scales and electrophysiology.
 5. **Biomarker discovery:** ER-stress/oxidative-stress fluid markers and neurofilament light as candidate progression biomarkers.
 6. **Genotype–phenotype curation:** systematically classify *REEP1* variants by mechanism (accumulation/gain-of-function → dHMN5B vs. loss-of-function → SPG31) to guide diagnosis and future allele-specific therapy.
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Report compiled from 8 confirmed findings and 34 reviewed papers. Evidence types span human clinical/genetic (small pedigrees, cohorts), model organism (mouse KI/KO, yeast, Drosophila), in vitro (iPSC, minigene, biochemistry), and ontology/database curation (MONDO, OMIM, HPO, Monarch).

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